Additional Exam Style Questions

Here are a collection of previous exam questions based on the material from weeks 1 to 5.

1. In a certain communication system, the noise introduced by a signal transmission channel is modelled as following a standard Laplace distribution, $X \sim \text{Laplace}(0, 1)$. The Laplace distribution is a continuous probability distribution with probability density function

$$f_X(x) = \frac{1}{2}e^{-|x|} = \begin{cases} \frac{1}{2}e^{-x} & x \geq 0 \\ \frac{1}{2}e^{x} & x < 0. \end{cases}$$

Here $|x|$ denotes the absolute value of x and $|x| = x$ for $x \geq 0$ and $|x| = -x$ for $x < 0$.

(a) Determine the values of θ for which the moment generating function is finite and show that the moment generating function of X, $M_X(\theta)$, is given by

$$M_X(\theta) = \frac{1}{1 - \theta^2}$$

when the moment generating function is finite. [6 marks]

(b) Hence use the moment generating function to determine the variance, $\mathbb{V}\text{ar}(X)$, of X when $X \sim \text{Laplace}(0, 1)$. [4 marks]

[Total 10 marks]

2. (a) Let X and Y be two independent random variables with respective moment generating functions given by:

$$M_X(\theta) = \frac{1}{1 - 3\theta}, \quad \text{if } \theta < \frac{1}{3}, \quad \text{and } M_Y(\theta) = \frac{1}{(1 - 3\theta)^2}, \quad \text{if } \theta < \frac{1}{3}.$$

Show that the moment generating function of $X + Y$ is given by:

$$M_{X+Y}(\theta) = \frac{1}{(1 - 3\theta)^3}, \quad \text{if } \theta < \frac{1}{3}.$$

[4 marks]

(b) Suppose that the revenue generated by a company in a week is a random variable Y, with moment generating function $M_Y(\theta)$ defined by:

$$M_Y(\theta) = \frac{1}{(1 - 258\theta)^3}, \quad \text{if } \theta < \frac{1}{258}.$$

Use the definition of moments to find the mean and variance of the random variable Y.

[5 marks]

[Total 9 marks]

3. Let $X_1, \ldots, X_n$ be a random sample from the probability density function defined by:

$$f_X(x) = \begin{cases} \lambda(x - \frac{1}{2}) + 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise} \end{cases}$$

where λ is an unknown parameter. Suppose that an estimator for λ is given by:

$$\hat{\lambda}(\underline{X}) = 12\bar{X} - 6.$$

(a) Find $\mathbb{E}[X]$ for this distribution. Hence, show that $\hat{\lambda}$ is an unbiased estimator of λ.

[6 marks]

(b) Find $\mathbb{E}[X^2]$ for this distribution and show that the mean squared error (MSE) of $\hat{\lambda}$ is given by

$$\text{MSE}(\hat{\lambda}) = \frac{12 - \lambda^2}{n}.$$

[7 marks]

(c) Is $\hat{\lambda}$ a good estimator as n approaches infinity? Discuss your answer.

[1 mark]

[Total 14 marks]

4. A medical researcher is studying the prevalence of a disease in a population, where the disease occurrence, X, follows a normal distribution whose mean and variance are equal, that is, $X \sim \text{N}(\theta, \theta)$, with mean θ and variance θ. The researcher takes a random sample of 3 patients, denoted by X_1, X_2, X_3, and aims to estimate the disease prevalence, θ, using two estimators:

$$\hat{\theta}_1 = \frac{X_1 + X_2 + X_3}{3}$$
$$\hat{\theta}_2 = \frac{X_1 + 2X_2 + 3X_3}{3}$$

(a) Which of the two estimators is unbiased. Justify your answer.

[5 marks]

(b) Determine the mean square error (MSE) for $\hat{\theta}_2$.

[3 marks]

(c) Given that the MSE of $\hat{\theta}_1$ is $\frac{\theta}{3}$, calculate the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ and determine which of the two estimators is more efficient. Provide reasons for your choice.

[2 marks]

[Total 10 marks]

5. A biologist is studying the distribution of a rare species of seed in a forest, specifically focusing on the proportion of seeds that germinate within a given time period. The researcher assumes that the success rate of germination for each seed, denoted as $x_1, x_2, \ldots, x_n$ follows a probability density function (pdf) defined by:

$$f_X(x) = \begin{cases} \lambda x^{\lambda-1} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

where $\lambda > 0$ is an unknown parameter related to the proportion of seeds that successfully germinate within the timeframe, and has to be estimated from the data.

(a) Derive the maximum likelihood estimator for λ.

[5 marks]

(b) Derive the method of moment estimator for λ.

[4 marks]

(c) Suppose three independent random samples obtained from the distribution $f_X(x)$ are: $0.1, 0.2,$ and 0.3. Which of the two estimators in **5(a)** and **5(b)** provides the smallest estimate for λ. Justify your answer

[2 marks]

[Total 11 marks]

6. Let $X_1, X_2, \ldots, X_n$ represent a random sample from the Gamma $(2, \beta)$ distribution with an unknown parameter β defined as:

$$f_X(x) = \begin{cases} \beta^2 x e^{-\beta x}, & x > 0, \\ 0, & \text{otherwise} . \end{cases}$$

(a) Derive the log-likelihood function for β.

[3 marks]

(b) Find the maximum likelihood estimator (MLE) for β. Verify that the log-likelihood is maximised.

[3 marks]

(c) Suppose that $x_1 = 0.13, x_2 = 0.21, x_3 = 0.18$ and $x_4 = 0.33$ are a random sample of size 4 from the distribution of X defined above. Calculate an estimate for β using the method of moment estimation (MME).

[4 marks]

[Total 10 marks]

7. A company is evaluating a new manufacturing process to see if it improves the diameter consistency of a component. A sample of 15 components produced with the new process has a mean diameter of $\bar{x} = 50.05$mm and a variance of $s^2 = 0.015\text{mm}^2$.

(a) Explain why

$$\frac{\bar{X} - \mu}{S^2/n}$$

is a pivotal quantity for $\bar{X}$, the mean diameter of the components produced by the new manufacturing process.

[3 marks]

(b) For the above data, calculate the 95% confidence interval for the mean diameter of the components produced by the new process. Does the calculated 95% confidence interval suggest that the new manufacturing process has a significant effect on improving the diameter consistency of the components compared to the target mean diameter of 50mm?

[3 marks]

(c) Using the χ^2-distribution, construct a 95% confidence interval for the variance of the diameter of the components produced by the new process. Determine if there is a significant difference in the variability of the component diameters compared to the population variance of 0.02mm^2.

[4 marks]

[Total 10 marks]

8. For a sample of 11 randomly selected patients, their blood glucose levels were measured using two different glucose meters. The data are presented in the following table.

Patient ID	1	2	3	4	5	6	7	8	9	10	11
Meter 1	60	88	73	61	49	69	93	71	94	66	54
Meter 2	62	88	75	61	48	72	90	74	93	63	55

A physician wants to determine whether the meters give significantly different readings. By constructing an appropriate 95% confidence interval, provide a conclusion to the physician's query. State any assumptions made.

[Total 8 marks]